# BIO-DATA / CURRICULUM VITAE PROFORMA

## DR. ANIKA SHARMA

Application for the post of Scientist 'E' — Avionics and Sensor Systems (deputation), Example Defence Research Laboratory (Advt. / Circular No. EDRL/RECT/2026/07)

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### Personal and Service Particulars (items 1–3)

| | |
|---|---|
| **1. Name and address (in block letters)** | ANIKA SHARMA |
| **2. Date of birth (in Christian era)** | |
| **3. (i) Date of entry into service** | 01/07/2016 |
| **3. (ii) Date of retirement under Central / State Government rules** | |

### 4. Educational Qualifications

| Qualification | Discipline / Subject | Institution / University | Year | Division / Grade |
|---|---|---|---|---|
| B.E. | Electronics and Communication Engineering | Northfield Institute of Science and Technology, Northfield University | 2004 | First Class with Distinction |
| M.Tech. | Avionics<br>Thesis: Fault-tolerant data acquisition for airborne test instrumentation | Coastal Institute of Aeronautical Engineering | 2010 | CGPA 8.6/10 |
| Ph.D. | Electrical and Electronics Engineering<br>Thesis: Lightweight deep networks for real-time enhancement of aerial imagery | Northfield Institute of Science and Technology | 2021 – ongoing | Ongoing |

### 5. Whether Educational and Other Qualifications Required for the Post Are Satisfied

| | Qualifications / experience required as mentioned in the advertisement / vacancy circular | Qualifications / experience possessed by the officer |
|---|---|---|
| Essential | Master's degree in Electronics / Electronics and Communication / Avionics Engineering from a recognised university. | M.Tech., Avionics — Coastal Institute of Aeronautical Engineering (2010) |
| Essential | At least 10 years of experience in design, integration or testing of avionics, embedded or sensor systems. | At least 23 years (since Aug 2002) in avionics, test & evaluation: Scientist E, Kestrel Atmospheric Instrumentation Laboratory; Senior Design Engineer, Halcyon Avionics Design Bureau; Graduate Engineer, Meridian Electronics Limited |
| Desirable | Ph.D. in Electronics, Electrical or Computer Engineering. | Ph.D., Electrical and Electronics Engineering — Northfield Institute of Science and Technology (ongoing) |
| Desirable | Experience in machine learning, computer vision or edge AI deployment for airborne or UAV platforms. | Skill: Image super-resolution; Skill: Object detection; Project: Edge super-resolution pipeline for UAV imagery |



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| | Qualifications / experience required as mentioned in the advertisement / vacancy circular | Qualifications / experience possessed by the officer |
|---|---|---|
| Desirable | Publications in peer-reviewed journals. | 6 peer-reviewed publication(s): Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs; An IoT telemetry framework for upper-air ground stations; A low-cost GNSS receiver chain for radiosonde tracking |
| Desirable | Experience of leading technical teams and project coordination. | Project: Radiosonde ground-station acceptance test protocol; Planned environmental qualification (MIL-STD-810 vibration, temperature and humidity) for line-replaceable units and documented the test results |
| Desirable | Design and evaluation of airborne sensor and avionics subsystems | Developed LabVIEW test sequences for power-supply and interface boards of avionics units; Mentored graduate engineer trainees on avionics bus protocols and test documentation; Project: IoT monitoring of laboratory infrastructure |
| Desirable | development of embedded processing for UAV payloads | Project: Edge super-resolution pipeline for UAV imagery; Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment; Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT |
| Desirable | preparation of technical specifications and test plans | Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules; Scientist E, Kestrel Atmospheric Instrumentation Laboratory |
| Desirable | guidance of junior scientists. | Publication: A low-cost GNSS receiver chain for radiosonde tracking; Scientist E, Kestrel Atmospheric Instrumentation Laboratory |

## 6. Statement on Essential Qualifications and Work Experience

**6. Please state clearly whether, in the light of the entries made above, you meet the requisite essential qualifications and work experience of the post**

## 7. Details of Employment, in Chronological Order

| Office / Institution | Post held on regular basis | From | To | Pay level / pay scale of the post | Nature of duties (in detail), highlighting experience required for the post applied for |
|---|---|---|---|---|---|
| Test Systems Group, Meridian Electronics Limited | Graduate Engineer | 08/2002 | 05/2004 | | Built automated test equipment for avionics line-replaceable units.<br>• Developed LabVIEW test sequences for power-supply and interface boards of avionics units. |



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| Office / Institution | Post held on regular basis | From | To | Pay level / pay scale of the post | Nature of duties (in detail), highlighting experience required for the post applied for |
|---|---|---|---|---|---|
| Avionics Integration Group, Halcyon Avionics Design Bureau | Senior Design Engineer | 06/2004 | 06/2016 | | Design, integration and environmental qualification of avionics line-replaceable units; development of automated test rigs; support to flight trials of UAV payloads.<br>• Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.<br>• Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.<br>• Planned environmental qualification (MIL-STD-810 vibration, temperature and humidity) for line-replaceable units and documented the test results.<br>• Mentored graduate engineer trainees on avionics bus protocols and test documentation. |



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| Office / Institution | Post held on regular basis | From | To | Pay level / pay scale of the post | Nature of duties (in detail), highlighting experience required for the post applied for |
|---|---|---|---|---|---|
| Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory | Scientist E | 07/2016 | Till date | Level 12 | Planning, testing and maintenance of radiosonde ground systems and upper-air instruments; management of the division workshop; leadership of IoT monitoring and AI/ML development; technical evaluation and procurement of instruments under GFR; guidance of project staff and interns.<br>• Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.<br>• Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.<br>• Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.<br>• Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.<br>• Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.<br>Additional charge: Workshop In-charge, Upper Air Division Workshop (04/2019 – Present)<br>Nodal officer: Nodal Officer, IoT and AI/ML Initiatives (01/2021 – Present) |

## Present Employment and Pay (items 8–15)

| | |
|---|---|
| **8. Nature of present employment, i.e. ad hoc, temporary, quasi-permanent or permanent** | Permanent |
| **9. If the present employment is held on deputation / contract basis: (a) date of initial appointment, (b) period of appointment on deputation / contract, (c) parent office / organisation, (d) name and pay of the post held in substantive capacity in the parent organisation** | |
| **10. If any post was held on deputation in the past: date of return from the last deputation and other details** | |
| **11. Present employer: Central Government, State Government, autonomous organisation, government undertaking, university or others** | |



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| | |
|---|---|
| **12. Whether working in the same department and in the feeder or feeder-to-feeder grade** | |
| **13. Whether in a revised scale of pay; if so, the date of revision and the pre-revised scale** | |
| **14. Total emoluments per month now drawn** | Pay level: Level 12 (7th CPC) |
| **15. If the employer does not follow Central Government pay scales: details of the latest salary slip (enclosed)** | |

## 16 A. Additional Information Relevant to the Post

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

Areas of specialisation: R&D in avionics systems (12 years); Edge AI for aerial imagery (5 years); Upper-air instrumentation (10 years); IoT monitoring systems (4 years).

## 16 A. Professional Training, Courses and Certifications

**Deep Learning for Edge Devices** 03/2022
Example Academy of Applied Computing
Used the model-compression techniques to deploy the drone super-resolution model on Jetson modules.
**Duration:** 40 hours
**Credential:** EAAC-DL-0042
**Valid until:** 03/2029
**Mode:** Online
**Skills:** Deep learning, Edge AI, TensorRT

**Workshop on Efficient Deep Learning for Embedded Vision** 12/2023
Example Conference on Instrumentation (ECI 2023)
**Duration:** 2 days

**Academic English proficiency test** *(Expired)* 05/2016
Example Language Testing Board
**Result:** Overall band 8.0
**Expired:** 05/2018

**Introduction to Machine Learning (12-week course)** 10/2018
Example Online University Consortium
**Duration:** 48 hours
**Result:** Elite (82%)
**Mode:** Online
**Skills:** Machine learning, Python

**Public Procurement and GFR 2017** 11/2019
Example Institute of Public Administration (sponsored by Kestrel Atmospheric Instrumentation Laboratory)
**Duration:** 5 days
**Skills:** Public procurement, GFR 2017, GeM

**National Seminar on Upper-Air Observing Systems** 08/2021
Example Society of Atmospheric Instrumentation
**Duration:** 1 day

**Mid-Career Training Programme for Scientists** 02/2020
Example Institute of Scientific Administration (sponsored by Ministry of Example Sciences)
**Duration:** 10 days


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### 16 B (i). Research Publications and Reports

[1] A. Sharma and R. Iyer, “Knowledge-distilled lightweight networks for drone image super-resolution,” accepted for publication in Example Transactions on Aerial Imaging.

[2] A. Sharma, K. Rao and R. Iyer, “Benchmarking embedded GPUs for real-time image enhancement,” Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] A. Sharma and R. Iyer, “Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs,” Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] A. Sharma, P. Rao and S. Gill, “An IoT telemetry framework for upper-air ground stations,” Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] A. Sharma and S. Verma, “Edge telemetry dashboards for upper-air ground stations,” Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] A. Sharma, P. Menon and R. Iyer, “A low-cost GNSS receiver chain for radiosonde tracking,” Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

### 16 B (i). Manuscripts Under Review / In Preparation

[1] A. Sharma, M. Khan and R. Iyer, “Detection-aware quality assessment of super-resolved drone images,” under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] A. Sharma, “Simulation study of balloon-released gliding sondes,” in preparation. *[In preparation]*

### 16 B (i). Special Projects

**Edge super-resolution pipeline for UAV imagery** *(Ongoing)* 09/2021 – Present

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

**Technologies:** NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

**Radiosonde ground-station acceptance test protocol** 01/2018 – 03/2019

Coordinator

Station acceptance testing varied between sites.

- A common acceptance protocol and checklist adopted for new ground-receiving systems.

**IoT monitoring of laboratory infrastructure** 02/2020 – 06/2022

Project lead

Instrument rooms lacked continuous environmental and power-quality monitoring.

**Technologies:** ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

- Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.
- Continuous monitoring with automated alerts replaced manual twice-daily readings.

### 16 B (ii). Awards, Scholarships and Official Appreciation

**Best Paper Award** 2023

Example Conference on Instrumentation (ECI 2023)

### 16 B (iii). Affiliation with Professional Bodies, Institutions and Societies

**Member, Institution of Example Engineers** 2010 – Present

Institution of Example Engineers



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### 16 B (vi). Other Information: Responsibilities and Committees

**Supervisor, M.Tech. interns in applied machine learning** 2022 – Present
Kestrel Atmospheric Instrumentation Laboratory
Supervise summer and dissertation interns on edge-AI projects.

**Chair, Technical Evaluation Committee for radiosonde procurement** 04/2023 – 02/2024
Kestrel Atmospheric Instrumentation Laboratory
Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

### 16 B (vi). Other Information: Talks and Presentations

**Invited talk: Edge AI for atmospheric observation** 2024
Example Symposium on Applied AI

### Basis of Application (items 17–18)

| | |
|---|---|
| **17. Whether applying on deputation (ISTC / STC), absorption or re-employment basis** | Deputation |
| **18. Whether belongs to SC / ST** | |

### Declaration by the Candidate

I have carefully gone through the vacancy circular / advertisement and I am well aware that the information furnished in the Curriculum Vitae duly supported by the documents in respect of Essential Qualification / Work Experience submitted by me will also be assessed by the Selection Committee at the time of selection for the post. The information / details provided by me are correct and true to the best of my knowledge and no material fact having a bearing on my selection has been suppressed / withheld.

Place: New Delhi
Date: (Signature)
**Anika Sharma**

### Certification by the Employer / Cadre Controlling Authority

The information / details provided in the above application by the applicant are true and correct as per the facts available on records. The applicant possesses the educational qualifications and experience mentioned in the vacancy circular. If selected, the applicant will be relieved immediately.

Also certified that:

(i) There is no vigilance or disciplinary case pending / contemplated against Anika Sharma.

(ii) The applicant's integrity is certified.

(iii) The APAR dossier in original is enclosed / photocopies of the APARs for the last five years, duly attested by an officer of the rank of Under Secretary to the Government of India or above, are enclosed.

(iv) No major / minor penalty has been imposed on the applicant during the last ten years / a list of major / minor penalties imposed during the last ten years is enclosed (as the case may be).

Place: (Signature)
Date: **Countersigned (Employer / Cadre Controlling Authority, with seal)**



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